



A display can only log the faults of up to eight PPC's, therefore the fault log range is selected by holding down the right arrow key and the Engineers Key (Fireye logo) simultaneously. The ranges are as follows;

Fault Mode on CANbus

0
1
2

Fault Log Range

No Logging
Address 0-7
Address 8-15

6.7.2 Startup

Problem	Possible cause	Suggested action
Burner will not start.	Control in non-volatile lockout. High control limit exceeded. Control in commission ratio mode. Burner off via serial comms. Waiting for burner controller.	Hold MUTE/RESET key when faults clear. Check EK18 and wait for press/temp to drop. Press NEXT to advance to next setpoint. Turn the burner ON using ComFire software (if fitted). Use section 4.2 and burner status (EK30) to see what the control is waiting for.
Drive positioning fault occurs before drives move to purge.	Stored close positions do not match actual close positions.	Check feedback potentiometers and motor micro-switches. Reset close positions using commission ratio mode.
Drive positioning fault occurs when drives reach purge.	Stored purge positions cannot be reached.	Check feedback potentiometers and motor micro-switches. Reset purge positions using commission ratio mode.
Drives stuck at ignition and burner has not fired.	Drives cannot reach ignition position.	Check motor micro-switches and linkages.

6.7.3 Commissioning

Problem	Possible cause	Suggested action
Cannot get past P0.	See 'burner will not start' above.	See 'burner will not start' above.
'Px' or 'Ax' display flashes constantly and motor positions cannot be altered.	Drives moving to position. Burner off in adjust ratio mode. A0, A1, A2 displayed. Waiting for burner controller.	Wait for drives to position. Turn burner on if you need to adjust points a3 onwards. Use section 4.2 and burner status (EK30) to see what the control is waiting for.
Option parameter not available.	Another option parameter must be set first.	Set option parameter (usually XX.0) to a non-zero value to enable other params. in group.
Option parameter not adjustable.	Wrong passcode entered or burner firing.	Enter the supplier passcode and/or turn the burner off before changing the value.
Not possible to enter commission ratio mode.	Supplier passcode incorrect. Burner firing (goes into adjust ratio mode).	Enter correct passcode. Turn burner off and try again.
Not possible to enter adjust ratio mode.	Supplier passcode incorrect.	Enter supplier passcode with the burner on.



6.7.4 Modulation

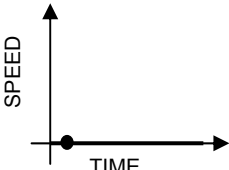
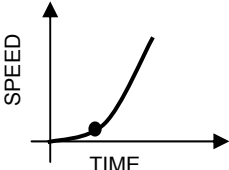
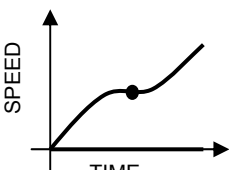
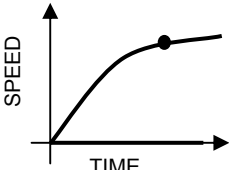
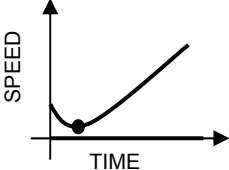
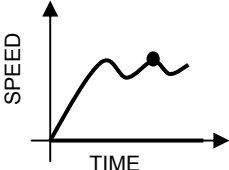
Problem	Possible cause	Suggested action
Sensor or modulation rate inaccurate.	V/I input setting incorrect. 5/24V supply setting incorrect.	Check option parameter. Check SENSOR IN and SUPPLY links.
Intermittent positioning faults.	Profile too close to micro-switch. Poor Ground (earth) or screening. Feedback potentiometer faulty. Communication to servo-motor(s).	Move end points further away from high/low. Check wiring. Move motor across range and check the feedback in commission ratio mode with the burner off. Check wiring.
Control stuck at low fire.	Meas. value exceeds setpoint. Warming limit active Control in manual. External low fire hold, or missing AUTO input from burner controller. In manual mode.	No fault. Wait for boiler to warm up. Check EK16. Press the AUTO key to change to auto mode Remove feed from Aux inputs (if selected). Check EK 6. Increase the mod. rate using 'UP' key. Check PID, proportional band must be greater than zero to modulate.
Control stuck at a modulation rate.	Serial communications. Control in commission mode.	Disable or change modulation slider in Computer software. Press RUN then ENTER to enter run mode.

6.7.5 Inverters



CAUTION

Due to the possibility of High Frequency electrical noise, **BOTH** ends of the shielded signal wire to the VSD **must** be grounded.

Problem	Possible Cause	Solution
	Inverter does not start because it does not receive a RUN signal.	Ensure that the inverter receives a RUN signal from the burner fan motor output on the PPC6000 series control.
	- Inverter has a slow start. - Late RUN signal. - Non-linear output from inverter or inverter's PID is enabled	- Ensure that the inverter's slow start feature is disabled. - Ensure that the inverter receives a RUN signal from the burner controller at the same time as the PPC6000. - Check that the inverter's output is selected to be linear, and that the inverter's own PID loop is disabled.
	- Current limit reached - Noise	- Slow down the inverter by increasing its acceleration / deceleration time settings. - Check cable screens.
	- Current limit reached. - Non-linear output from inverter or inverter's PID is enabled.	- Slow down the inverter by increasing its acceleration / deceleration time settings. - Check that the inverter's output is selected to be linear, and that the inverter's own PID loop is disabled.
	Fan failed to stop before restart.	Increase the inverter stop time by increasing option parameter 9.3 on the PPC6000 series control.
	Control is unstable	- Adjust option parameters 9.0, 9.2 and 9.3 on the PPC6000 series control to reduce accuracy & slow down control response. - Check Option parameter 9.4 matches the acceleration / deceleration time programmed into the VSD. - VSD is current limiting. Increase acceleration / deceleration time in VSD and option parameter 9.4.



In extreme cases, it may be necessary to increase the inverter error tolerance to prevent non-volatile lockouts caused by positioning faults (set option 9.1 = 1). **This must only be changed if an inverter error of ± 55 will not cause unsafe combustion.**

Variable Speed Drive Selection:

Important Note: When selecting a VSD or variable frequency drive (VFD) consideration must be given to the size and weight of the blower wheel. While a “fan” application is by nature a variable torque load, the inertia (weight) of the blower wheel must be considered. Typical HVAC type drives do not speed up or slow down large blower wheels with enough control to operate within the tolerance of the PPC or NX series control. Therefore, a VSD (VFD) with some built in breaking capability, such as a constant torque drive, should be used. At a minimum, if the drive has a constant torque setting, it must be enabled to minimize faults due to drive hunting.

Problem	Possible cause	Suggested action
Oxygen display not available.	No oxygen probe interface unit serial number entered.	Enter serial number into option parameter 30.0
No inlet temperature display	No inlet temperature unit serial number entered.	Enter serial number into option parameter 35.0
Inlet or flue temperature display flashes 'Hi'	Inlet air temperature sensor / O2 probe not installed and wired correctly.	Check wiring.
No efficiency display or efficiency incorrect.	No oxygen display. No inlet temperature display. No calorific value for the current fuel been entered into 35.X?	See 'No oxygen display' See 'No inlet temperature display' Enter the value for the fuel in use.
Oxygen display shows '---'	Probe not heated up yet, or other probe fault.	Check EK45. Check EK73 – must be above 600°C to work.
Oxygen trim will not work	Trim is not operative. Boiler just fired up. Probe calibration values not entered correctly. Probe not heated up yet. Probe installed incorrectly. Set to monitor only. Probe in calibration. Trim being reset. Trim limits set to 0.0% of flow. Trim integral gain set to zero. Commissioning data missing.	Use EK45 and 46 to decide if trim is operative. Check that option 30.5 = 1. Check option parameter 31.1 to 31.4. Wait for the boiler transport time (after ignition). Wait for modulation. Re enter values (options 20.1,20.2) Check EK75 – must be above 600°C to work. Check wiring. Check option parameter 30.5 set to 1. Check option parameter 30.6 set to 0. Check option parameter 30.8 set to 0. Check option parameters 32.X. Check option parameters 33.X are non zero. Check oxygen and flow values been entered for all profile points in the firing range.



6.8 YB110/TSD104 Fault Messages

Due to the increased use of the Nexus Touchscreen in combination with the BurnerLogix, the numbering format for fault messages is formatted as below. The fault numbers are taken from table 7 in the BurnerLogix manual BL-1001-pdf, but with 200 added to the message number. The 200 is added because there are codes associated with the PPC6000 that utilize the lower levels codes. The list of touchscreen alarm codes associated with the BurnerLogix is listed below:

Fault Number	Description
201	T16 M-D LIMIT OPEN-AUTO
202	HOLD FALSE FLAME-STANDBY
203	LOW FIRE PURGE
204	HOLD M-8 LIMIT OPEN-PURGE
205	AIR FLOW OPEN
206	T16 M-D LOW LIMIT-AUTO
207	LOCKOUT FLAME FAIL-PTFI
208	T16 INPUT CLOSED
209	HOLD M-D LIMIT OPEN
210	IGNITION TIMING-PTFI
211	CHECK FLAME SIGNAL-MTFI
212	FLAME SIGNAL - AUTO
213	CYCLE COMPLETE-POSTPURGE
214	STANDBY L1-3 OPEN
215	T21 INPUT CLOSED
216	LOCKOUT M-D LT OPEN-PTFI
217	LOCKOUT LT M-8 LIMIT OPEN
218	LOCKOUT LIMIT M-D OPEN
219	LOCKOUT FLAME FAIL - MTFI
220	LOCKOUT FALSE FLAME
221	LOCKOUT 3-P INTLK OPEN(P)
222	LOCKOUT 3-PINTLK CLOSED
223	HOLD 3-P INTLK CLSED STBY
224	HIGH FIRE PURGE
225	LOCKOUT M-D LT OPEN-MTFI
226	LOCKOUT 3-P INTLK O AUTO
227	LOCKOUT 3-P INTLK O MTFI
228	LOCKOUT 3-P INTLK O PTFI
229	LOCKOUT T13 FVES OPEN
230	LOCKOUT 3-P INTLK O - PP
231	CHECK FLAME SIGNAL-PTFI
232	CHECK M-8 HI LIMIT (HFP)
233	CHECK M-D LOW LIMIT (LFS)
234	FLAME SIGNAL - PTFI
235	HOLD T21 OR CHK LF AUTO
236	FLAME SIGNAL - MTFI
237	LOCKOUT FLAME FAIL AUTO
238	HOLD 3-P INTLK O - PP
239	LOCKOUT FUELVOLVE ST CH
263	HIGH WATER
264	LOW WATER
265	HIGH GAS PRESSURE
266	LOW GAS PRESSURE
295	AUX GAS



Fault Number	Description
267	LOW OIL PRESSURE
296	HIGH OIL TEMPERATURE
268	LOW OIL TEMPERATURE
269	LOW ATOMIZING MEDIA
270	HIGH PRESSURE
271	HIGH TEMPERATURE
272	AUX #4 OPEN
273	AUX #5 OPEN
274	AUX #6 OPEN
275	AUX #7 OPEN

Interlock annunciation messages THESE CAN BE MODIFIED BY THE USER

Fault Number	Description
241	HIGH WATER
242	LOW WATER
243	HIGH GAS PRESSURE
244	LOW GAS PRESSURE
292	AUX GAS
245	LOW OIL PRESSURE
256	HIGH OIL TEMPERATURE
246	LOW OIL TEMPERATURE
247	LOW ATOMIZING MEDIA
248	HIGH PRESSURE
249	HIGH TEMPERATURE
250	AUX #4 OPEN
251	AUX #5 OPEN
252	AUX #6 OPEN
253	AUX #7 OPEN
240	AIR FLOW OPEN
294	OP CONTROL OPEN
260	L1-3 AUX #1 OPEN
261	L1-3 AUX #2 OPEN
262	L1-3 AUX #3 OPEN

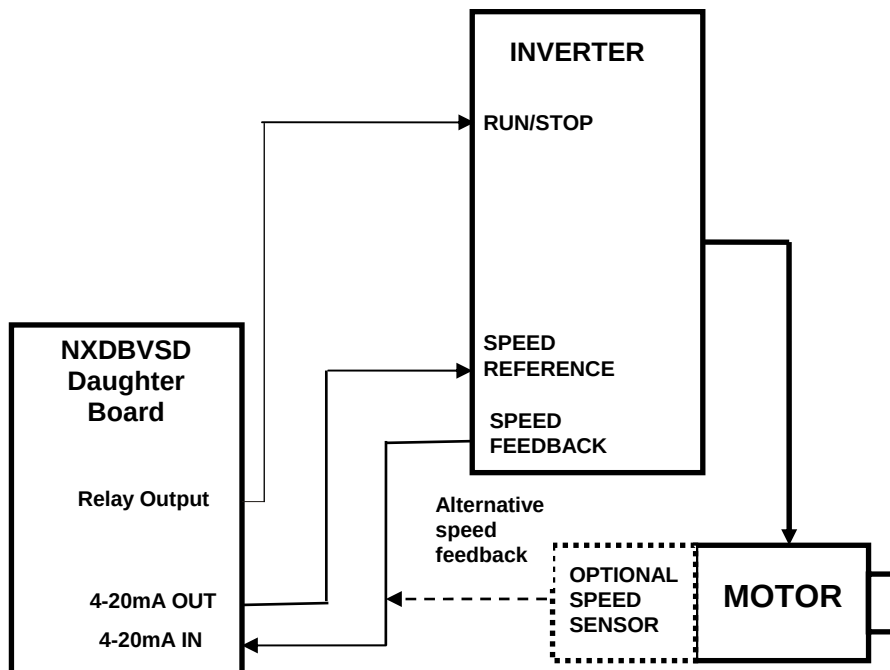
7 Inverter (variable speed) drives

This manual may cover more than one model in the PPC6000 series. Check for additional information at the end of this chapter.

7.1 System configuration

The PPC6000 control can control one or two inverter drive(s) to control a fuel and/or air drive(s) using a daughter board (fitted within the PPC6000 enclosure). The daughter board controls the inverter(s) using 4-20mA output channel(s). The output(s) are the speed reference signal to the inverter(s). A 4-20mA or encoder input is used to monitor the feedback signal from each inverter, representing the speed of the motor(s).

In addition to the 4-20mA signals, the inverter(s) also require motor start signal(s) that can be taken from the volt-free relay contact(s) on the daughter board, please ensure its function is set correctly using the relevant option parameter.





7.2 Description of control method

During start-up of the burner, and before any drives move to purge, the PPC6000 control, moves all drives to their minimum positions by driving servo-motors down and setting an output current of 4mA to each inverter drive. The control waits for the inverter stop time Option 9.3 and then monitors the feedback signals of all drives and compares them to the values which were stored in memory at commission time. If the values do not match those stored in memory, a positioning fault is given and the control performs a non-volatile lockout.

This test proves the feedback from each drive is operating correctly and that the potentiometer (where fitted) has not moved on the servo-motor shaft.

If the test is successful, the control moves all selected drives to their commissioned pre-purge positions (outputs 20mA to all inverters). If the drives cannot achieve their pre-purge positions as set during commissioning, a positioning fault is given and the control performs a non-volatile lockout.

If the burner start-up is successful, the control will control the frequency of the inverter during modulation. A closed loop method is employed, where adjustments to the speed reference signal are made based on the value of the feedback signal.

If the feedback signal from the drive is lost during a run condition (e.g. the loop current drops below 3.5mA or encoder feedback fails) the control will perform a non-volatile lockout.

7.3 Setting up the inverter for use with the PPC6000 series

The PPC6000 will work with most inverters that meet the following criteria:

- 4-20mA speed reference input.
- 4-20mA speed feedback output or use separate encoder.
- Remote run/stop command.
- The inverter must be powerful enough to accelerate / decelerate the motor as required. A good industrial inverter will have a significant overload capability meaning that it can supply well above the rated motor load for some time. This might be something like 150% (or more) for 1 minute. An HVAC inverter typically will have no (or very limited) overload capability and may not be able to accelerate / decelerate the motor quickly without current limiting. This can cause drive position faults when driving to purge.

Analog inputs and outputs:

The analog inputs and outputs (i.e. the 4-20mA reference and feedback signals) on the inverter should be configured in the following way:

- 4-20mA signal (this may be an option parameter and/or a board jumper)
- Speed reference input
- Speed reference output (unless using encoder)
- Minimum frequency 0Hz for 4mA signal
- Maximum reference frequency as required (nominally 60Hz) for 20mA signal
- Maximum feedback frequency same as reference for 20mA signal (unless using encoder)
- No filtering (time constant = 0.0) and no rate limiting

Digital input:

The digital input (i.e. the RUN/STOP) on the inverter should be configured in the following way:

- RUN/STOP from external input (NOT inverter keyboard)
- RUN/STOP active high (i.e. energize to start)

Control characteristics:

The control characteristics of the inverter should be configured in the following way:

- Acceleration and deceleration time approximately 25 - 40 seconds, must be the same.
- Straight line (linear) acceleration between reference points
- Motor to coast to a stop when RUN signal is removed
- No critical frequencies
- DC braking may be needed if motor deceleration is not linear

Motor characteristics:

The following motor characteristics should be entered into the inverter's option parameter list:

- Motor nominal voltage, power, current and frequency (see motor plate and/or supplier data)
- Motor current and temperature limits
- U/F ratio. Use the 'SQUARED' option for fans and pumps.
- Motor slip ratio. Use drive default, if unknown

7.4 Setting up the PPC6000 series for use with a inverter drive

In order to use a inverter drive with a 4-20mA interface, the optional inverter interface daughter board must be fitted. The following steps must then be taken:

- Enter commission mode.
- Enter option set mode.
- Look at the display for the inverter output. The display should show 0 for a feedback signal of zero speed (4mA) and 999 for a feedback signal of 20mA. If using an encoder feedback, the value displayed will depend on the frequency measured by the input on the daughterboard. See note below. If the display is flashing 'High', the feedback signal is less than 3.5mA or more than 21.0mA, the polarity of the wiring is reversed or the encoder is faulty. In all these cases, check the wiring and/or the option parameters on the inverter.
- Monitor the reference signal from the PPC6000 daughter board. With the inverter drive selected, the UP key should increase the current gradually up to 20mA and the DOWN key should decrease the current gradually down to 4mA.
- For encoder feedback signals, run the inverter to its maximum value (normally 50Hz or 60Hz, depending on motor rating / local mains frequency). Look at engineers key 69 or 70 (depending on VSD1 or VSD2). This will give you a measured pulse frequency in Hertz. Add about 2% to 5% to this value (to ensure it is the maximum you would ever expect to get) and enter this value into option parameter 9.5 or 9.6. The displayed value for this drive should now be 950 to 990 (i.e. 95% to 99%). For reliability it should be ensured that the feedback can't go over 99.9% (999). This is why a small percentage is added to the 100% scale value entered in the option parameter. See option parameter 9.5 for a way to check this value using a calculation.
- For the rest of the commissioning procedure, treat the inverter drive in the same way as a motor. The inverter drive has a closed position (4mA), a maximum position (20mA), a pre-purge position, an ignition position and up to 20 profile positions.

8 The oxygen trim option



CAUTION

- Use extreme care when handling the oxygen probe and wear heatproof gloves.
- Ensure the burner is off before removing the oxygen probe from the flue.
- If the boiler is to be operated with the probe removed, fit the blanking cover supplied since dangerous levels of carbon monoxide may be present in the flue.

8.1 Oxygen Pro probe description

The oxygen trim / monitoring function is designed to be used with an NX2012 oxygen probe. This probe offers fast, accurate response and good reliability when mounted in accordance with the guidelines in this section. The probe is available in three different sizes.

8.1.1 Installation of oxygen probe

8.1.2 Mounting the oxygen probe

